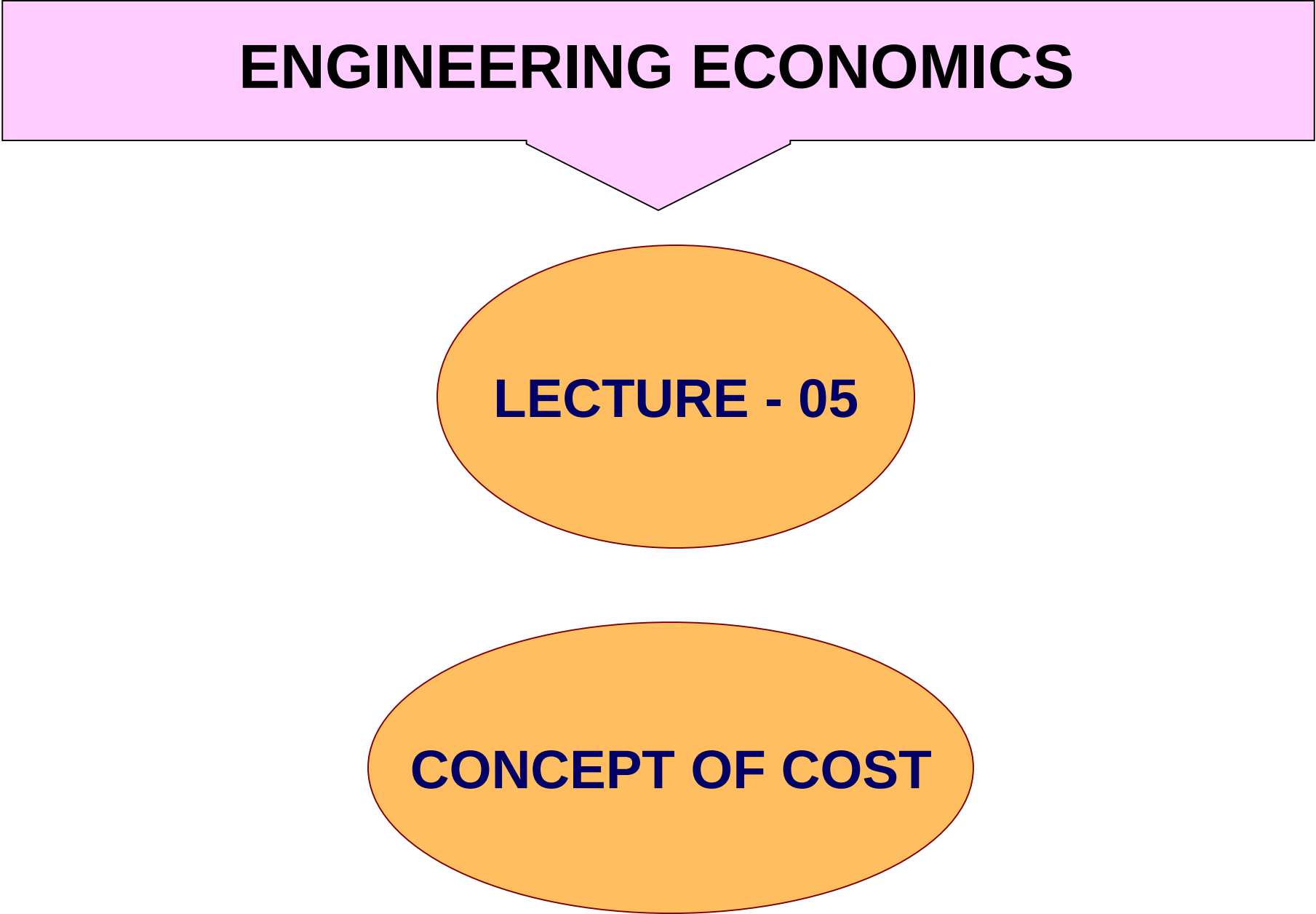


ENGINEERING ECONOMICS



LECTURE - 05

CONCEPT OF COST

First cost is the initial cost of capitalized property, including transportation, installation and other initial expenditures.

For **purchased equipment**, these include the purchase price plus shipping cost, installation cost, and training cost.

For a **fabricated structure, system or equipment**, they include engineering design and development cost, test and evaluation cost, and construction or production cost as well as shipping, installation and training costs.

Operation and maintenance cost is that group of cost experienced continually over the useful life of the activity.

Included in this category are labor costs for operating and maintenance personnel, fuel and power costs, operating and maintenance supply costs, spare and repair part costs, cost for insurance and taxes, and a fair share of indirect costs called overhead or burden.

These costs can be substantial and often they exceed the first cost in total amount.

The timing of their occurrence differ substantially, however, in that operating and maintenance costs occur over time until the structure, system or equipment is retired from service.

Direct costs can be reasonably measured and allocated to a specific output or work activity.

i-e labor and material directly allocated with a product, service or construction activity

Indirect costs are difficult to allocate to a specific output or activity.

i-e costs of common tools, general supplies, and equipment maintenance

NOTE: Indirect costs, Overhead and Burden are the same

Fixed costs is that group of costs involved in a going activity whose total will remain relatively constant throughout the range of operational activity.

Typical fixed costs include taxes, administrative expenses, license fees etc

The concept of fixed cost has a wide application. Almost any task involves preparation independent of its extent. Thus to paint a small area may require as much effort for the cleaning of a brush as to paint a large area.

Similarly, manufacturing involves fixed costs that are independent of the volume of output.

In practice fixed costs are only relatively fixed, and their total may be expected to increase somewhat with increased activity.

Like when plant expansion or shutdown is involved fixed costs will be affected.

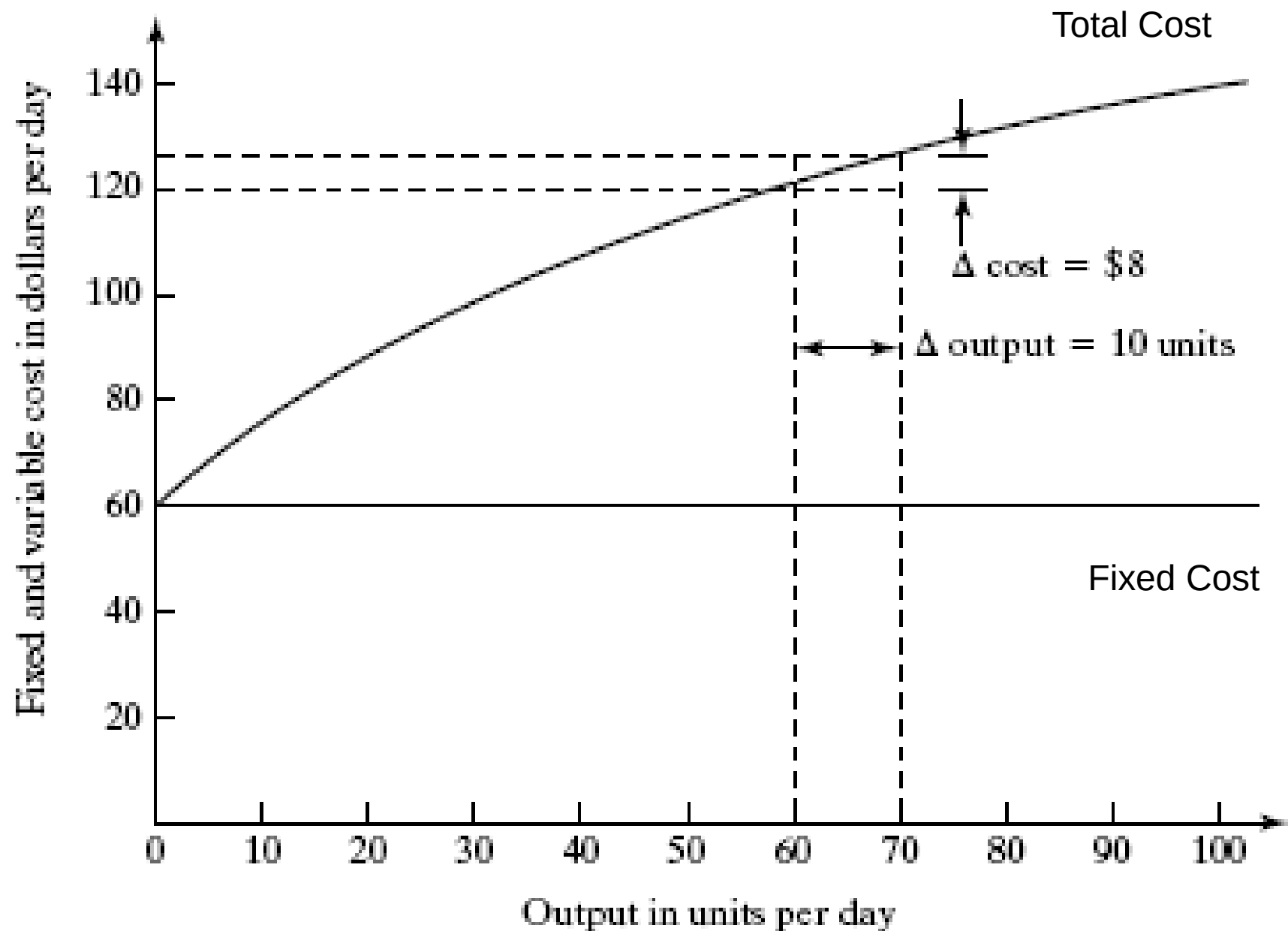
Variable costs are that group of costs that vary in some relationship to the level of operational activity.

For example the amount of paint used may be expected to be proportional to the area painted.

In manufacturing, the amount of material needed per unit of product may be expected to remain constant and, therefore, the material cost will vary directly with the number of unit produced. In general, all costs such as direct labor, direct material, which can readily be allocated to each unit produced are considered to constitute variable costs.

Incremental cost is the additional cost that results from increasing the output of a system by one more unit.

Reference is usually made to an increase of cost in relation to some other factor, thus resulting in such expressions as incremental cost pr ton, incremental cost per gallon, or incremental cost pr unit of production.



The figure illustrates the nature of fixed and variable cost as a function of output in units. The incremental cost of producing 10 units between outputs of 60 to 70 units per day is illustrated to be \$8. thus the average incremental cost of these 10 units may be computed as $\Delta \text{cost} / \Delta \text{output} = \$8/10 = \$0.80$ per unit.

A sunk cost is a past cost that can not be altered by future action and is therefore irrelevant.

Most decision makers seek the course of action that is expected to result in the most favorable future benefits. Because only the future consequences of investment alternatives can be affected by current decisions, an important principle in economy studies is to disregard cost incurred in the past.

Recurring costs are repetitive and occur when a firm produces similar goods and services on a continuing basis.

- Variable costs are recurring costs because they repeat with each unit of output .
- A fixed cost that is paid on a repeatable basis is also a recurring cost:
 - Office space rental

Nonrecurring costs are those that are not repetitive, even though the total expenditure may be cumulative over a relatively short period of time.

- Typically involve developing or establishing a capability or capacity to operate.
- Examples are purchase cost for real estate upon which a plant will be built, and the construction costs of the plant itself.

CASH COST AND BOOK COST

- Cash cost is a cost that involves payment in cash and results in cash flow.
- Book cost or noncash cost is a payment that does not involve cash transaction; book costs represent the recovery of past expenditures over a fixed period of time.

DEPRICIATION

- Depreciation is the most common example of book cost; depreciation is what is charged for the use of assets, such as plant and equipment; depreciation is not a cash flow.

Life Cycle Cost (LCC)

LCC is defined as all costs, both non recurring and recurring, that occur over the life cycle.

During the acquisition phase, non recurring costs are incurred, and these constitute the first cost of the structure or system.

During utilization phase, recurring costs are experienced. Detail of LCC is under these headings:

- **Cost Generated over the Life Cycle**

- **Cost Commitment over the Life Cycle**

INCURRED COST

- Incurred expenses have been charged or billed but are not yet paid.
- In other words, an expense incurred is **the cost when an asset is consumed**. A paid expense has been paid off by the company.

COMMITTED COST

- Committed Costs are fixed or budgeted or confirmed payments to be made in future to the vendors for goods or services to be taken, which are necessary for smooth flow of the business and the absence of which may disrupt the main operations of a business which may have credible impact on the company. Examples include rent cost of office, rental costs of factories, fixed pay for employees, retirement benefits paid to employees,

COMMITTED COST

- A committed cost is **an investment that a business entity has already made and cannot recover by any means**, as well as obligations already made that the business cannot get out of.
- One should be aware of which costs are committed costs when reviewing company expenditures for possible cutbacks or asset sales

EXAMPLES OF COMMITTED COST

Particulars	Amount (\$)
Fixed Pay of Employees	1250000
Factory Rent	350000
Warehouse Rent	125000
AMC for Plant & Machinery	75000
Insurance	45000
Audit Services	125000
Consumables Cost	750000

What is Life Cycle Cost?

- The life cycle cost (LCC) of any eqpt is the total “lifetime cost to purchase, install, operate, maintain, and dispose of that eqpt**
- The components include initial cost, installation and commissioning costs, energy costs, operation costs, maintenance and repair costs, down time costs, environmental costs, and decommissioning and disposal costs.**
- Some studies have shown that 30% to 50% of the energy consumed by pump systems could be saved through equipment or control system changes.**

Cost Generated over the Life Cycle

Many systems and products are planned, designed, produced and operated with very little concern for their life cycle cost.

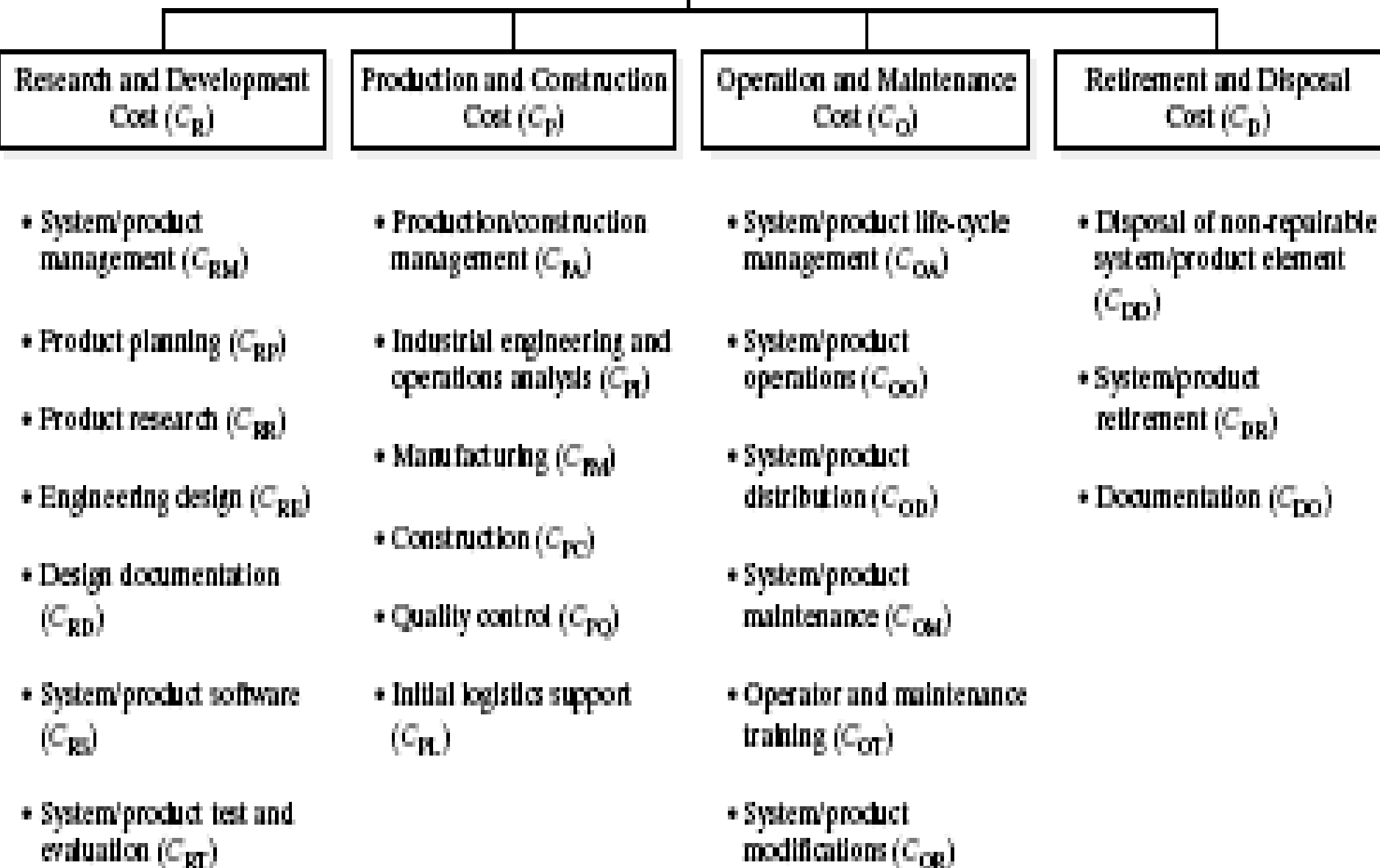
Although different aspects of cost have been considered in the development of new systems these costs have often been viewed in a fragmented manner.

The costs associated with activities such as research, design, testing, production or construction, consumer use and support have been isolated and addressed at various stages in the product life cycle, and not viewed on an integrated basis.

In general the elements of life cycle cost fall into categories that are based on organizational activity over the life cycle. These are illustrated in the figure below.

- See details on next slide

**Total System/
Product Cost (C)**



Experience has indicated that a large portion of the total cost for many systems is the direct result of activities associated with their operation and support, while the commitment of these costs is based on decisions made in the early stages of the system life cycle.

Further, costs associated with the different phases of the life cycle are interrelated.

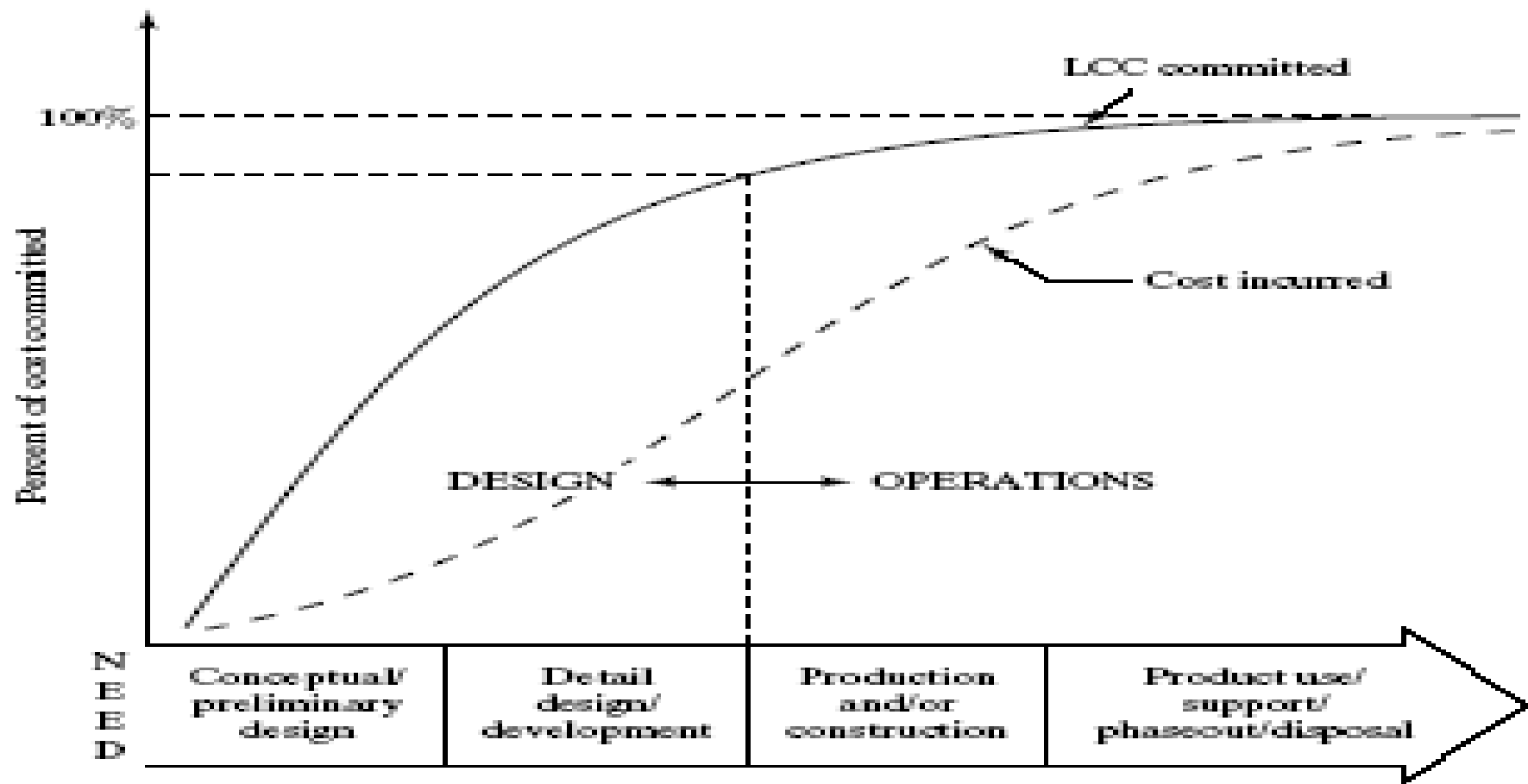
Thus in addressing the economic aspects of a system one must look at total cost in the context of the entire life cycle.

Cost Commitment over the Life Cycle

Throughout the system/product life cycle there are many actions required. The majority of the actions, particularly those at the earlier stages, have life cycle implications and definitely affect life cycle cost.

Life cycle cost and economic analysis should originate early in the product life cycle during conceptual and preliminary design.

The figure illustrate a characteristic cumulative life cycle cost curve related to action occurring during the various phases of the life cycle.



As illustrated more than half of the projected life cycle cost is committed by the end of the system planning and conceptual design even though actual expenditures are relatively minimal by this point in time.

Discussion